Simulation Arena

# Discount Leakage & Cherry-Picking: The Basket Sabotage

**Cherry-pick risk 71/100 (Moderate): give this discount and there is a 87.0% chance the Support line or another offsetting line disappears, costing $188k - 38.5% of the deal's profit.**

Seed 42 | Run run_1c41df28a020 | 2,000 scenarios tested | Cluster bootstrap resampling with empir… | 21 Sep 2026, 22:33

Same seed, same numbers - this run can be reproduced exactly.

## How this works

We went back through your past multi-line deals and found the ones where a line was discounted past 30%. In those deals we measured how often the other lines were cancelled inside 6 months, and compared it with how often those lines die anyway. That difference is the cherry-pick signal. We then rebuilt this quote 2,000 times, each time re-drawing whole past deals so the answer reflects how much history we actually have, and re-priced the basket with whatever survived. The risk score is three things you can check by hand: how far past your discount limit the ask sits (25%), how likely the offsetting lines are to vanish (50%), and how much of the profit sits in the line being cut (25%).

## The numbers that matter

CHERRY-PICK RISK SCORE
**71 / 100**
Built from three checkable pieces; the biggest one is how likely the offsetting lines are to vanish.
vs Moderate risk

PROFIT LIKELY TO LEAK AWAY
**$188k**
The profit this deal loses once you allow for how buyers actually behave.
vs 38.5% of the deal's profit

CHANCE THE SUPPORT LINE IS DROPPED
**64.0%**
How often the line that pays for the discount dies in the first 6 months.
vs 2.9x the normal 22.0%

CHANCE ANY OFFSETTING LINE DIES
**87.0%**
The odds the basket does not survive intact.

PROFIT EXPOSED ON THE SUPPORT LINE
**$96k**
$150k of profit with a 64.0% chance of walking.

PROFIT AT RISK
**38.5%**
Share of the bundle's profit that is exposed to cherry-picking.

PROFIT IF THEY KEEP EVERYTHING
**$489k**
The spreadsheet number: 54.3% margin on $900k.

PROFIT YOU SHOULD ACTUALLY PLAN FOR
**$301k**
The honest number after cherry-picking.
vs as low as $0 in a bad run

CHANCE THE BUNDLE ENDS UNDER YOUR 35% FLOOR
**14.2%**
How often the surviving basket is not worth signing at all.

## How we got there

### 1. What is actually on the table

The basket is worth $900k and makes $489k if nothing is dropped. But 61.3% of that profit sits in Software - the exact line the buyer wants 35% off, which is 5 points past the discount line you set yourself.

Evidence: $300k of $489k profit in Software

### 2. What history says happens next

You gave us no history to learn from, so we used your own numbers: 22.0% of lines die anyway, and 64.0% die once a deep discount is given.

Evidence: source: the rates you typed in

### 3. What the rebuild shows

Rebuilding this quote 2,000 times, the profit lands at $301k on average, and in an unlucky run of twenty it is $0 or worse. There is a 14.2% chance the bundle that survives is worth less than your 35% margin floor.

Evidence: $188k average shortfall

### 4. What to do about it

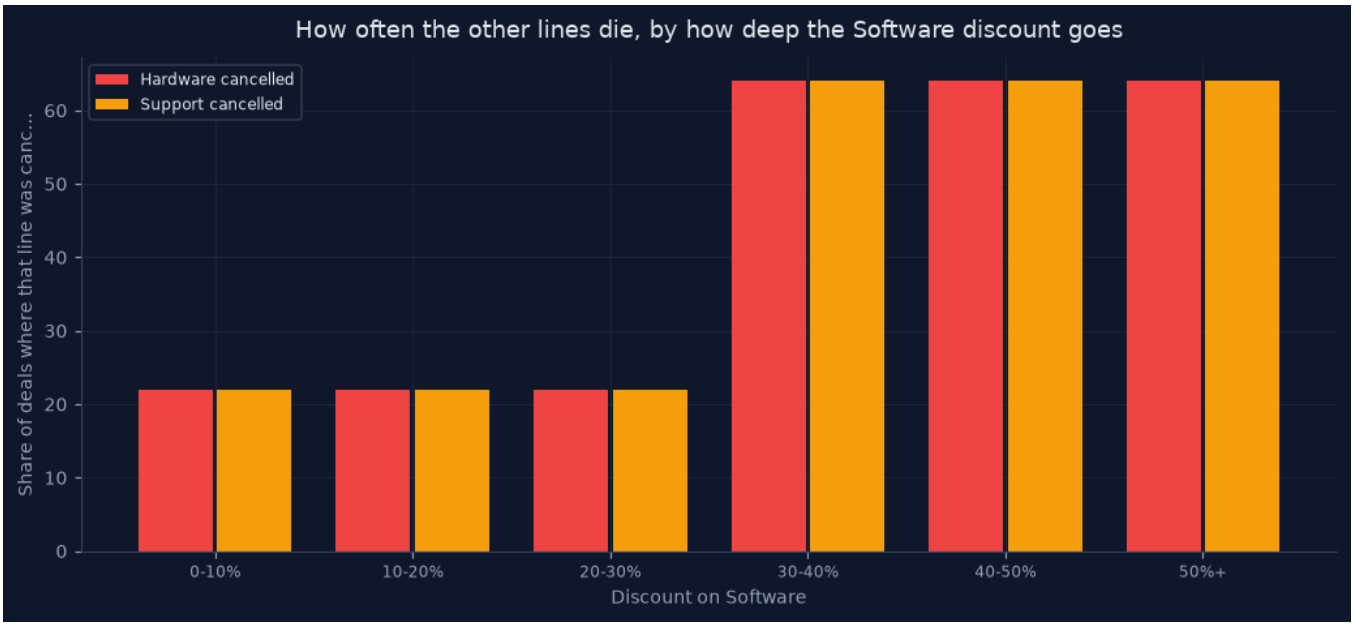
The biggest driver of the score is how likely the offsetting lines are to vanish. Tie the discount to the behaviour: if the Support line is not live at 80% of committed hours by month 6, the 35% Software discount comes off and Software reverts to standard pricing - worth up to $210k back.

Evidence: clawback up to $210k

## Charts

### How often the other lines die, by how deep the Software discount goes

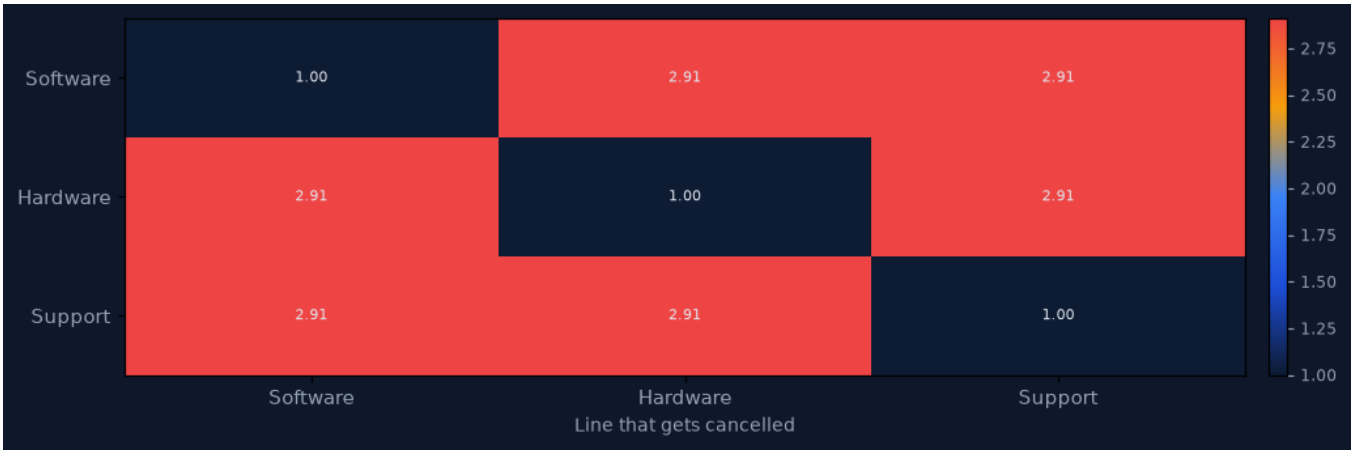
No history supplied - this is your assumption, drawn as a step




*What this shows: If the bars jump once the discount passes your limit, buyers are not just buying more - they are buying differently. That jump is the cherry-pick signal.*

### Which discount poisons which line

Numbers are 'times more likely than usual' when the row's line is discounted past 30%




*What this shows: Anything above 1.0 means the two events travel together. The hottest cell is the pair you should write a contract clause about.*

### What this deal actually earns

2,000 rebuilt bundles - the dashed line is the number on the quote




*What this shows: The gap between the tall bars and the 'if nothing is dropped' marker is the cost of cherry-picking. The left tail is the bad case you are signing up for.*

# Tables

## The quote, line by line

Where the profit sits, and how exposed each line is.

| Line | List price | Discount | Price after discount | Your margin | Profit | Share of profit | Chance it is dropped | Profit at risk |
|---|---|---|---|---|---|---|---|---|
| Software | $600,000 | 35.0% | $390,000 | 85.0% | $300,000 | 61.3% | 22.0% | $66,000 |
| Hardware | $300,000 | 5.0% | $285,000 | 18.0% | $39,000 | 8.0% | 64.0% | $24,960 |
| Support | $250,000 | 10.0% | $225,000 | 70.0% | $150,000 | 30.7% | 64.0% | $96,000 |

Row 1 carries the discount being asked for.

## What happens to each line when another line is discounted past 30%

Read across a row: 'when THIS line is deeply discounted, the chance THAT line dies'. A figure above 1.0 means the two events travel together.

| Discounted line (A) | Line that dies (B) | Deals like this | Normal rate | Rate after the discount | Times more often | Link strength |
|---|---|---|---|---|---|---|
| Software | Software | - | 22.0% | 22.0% | 1.00x | - |
| Software | Hardware | - | 22.0% | 64.0% | 2.91x | assumed |
| Software | Support | - | 22.0% | 64.0% | 2.91x | assumed |
| Hardware | Software | - | 22.0% | 64.0% | 2.91x | assumed |
| Hardware | Hardware | - | 22.0% | 22.0% | 1.00x | - |
| Hardware | Support | - | 22.0% | 64.0% | 2.91x | assumed |
| Support | Software | - | 22.0% | 64.0% | 2.91x | assumed |
| Support | Hardware | - | 22.0% | 64.0% | 2.91x | assumed |
| Support | Support | - | 22.0% | 22.0% | 1.00x | - |

No history was available - these are your assumptions, not evidence.

## What the bundle actually earns

Every one of the 2,000 rebuilt bundles, summarised.

| Scenario | Profit | Margin | Versus the plan |
|---|---|---|---|
| If they keep everything | $489,000 | 54.3% | the number on the quote |
| Most likely outcome | $300,000 | 33.3% | -$189,000 |
| A bad run (1 in 20) | $0 | 0.0% | -$489,000 |
| Worst bundle we saw | $0 | 0.0% | -$489,000 |

Row 2 is the number to plan with.

## What we recommend

**Approve the 35% only as a conditional discount - expect to lose $188k if you do not tie it to the Support line.**

**Risk Moderate** 71 out of 100 | Confidence: Low

### Why

• 61.3% of this deal's profit sits in Software, the line being cut by 35%.

• Past deals like this one lose the offsetting lines 87.0% of the time (the rates you typed in).

• Expected profit after cherry-picking is $301k, not $489k - a $188k gap.

### What to do

**1.** Make the 35% Software discount conditional: if the Support line is not live at 80% of committed hours by month 6, Software pricing reverts to the standard tier (recovering up to $210k).

**2.** Do not let them drop one offsetting line and keep the other: make the discount conditional on the whole Hardware, Support basket surviving, not on any single line.

**3.** Put the test at month 6, not at signature.

**4.** There is a 14.2% chance the surviving bundle sits under your 35% floor - require CF sign-off above 30% on any single line.

**5.** Upload past quote lines with discount and cancellation flags so this number comes from evidence instead of your assumption.

**The trade-off:** Walking away from the 35% risks the deal; giving it unprotected costs about $188k. The clause lets you keep the deal and cap the loss. If the buyer refuses any condition, the mildest version is to cap Software at 30%, which still protects $30k.

## What we assumed and what to watch

### Worth knowing before you act

• No usable history was mapped (we need a category, the discount given and whether the line was cancelled), so every probability here is your typed-in assumption: 22.0% normal abandonment, 64.0% after a deep discount.

### Assumptions

• Quote under review: Software $600,000 at 35% off (85.0% margin), Hardware $300,000 at 5% off (18.0% margin), Support $250,000 at 10% off (70.0% margin).

• Deep discount means anything past 30%; a line counts as abandoned if it dies within 6 months.

• Abandonment rates come from the rates you typed in: 22.0% normal, 64.0% after a deep discount.

• Risk score = 25% x how far past your discount limit the ask sits (10 points past = full marks) + 50% x chance an offsetting line dies + 25% x share of profit sitting in the discounted line.

• A cancelled line is treated as fully lost: no revenue and no cost.

• Bundle margin floor 35%; minimum support commitment 80% of hours.

• We replayed 2,000 bundles built from your own past deals; seed 42 - re-running gives identical numbers.

- Quote / deal ID -> not in your file; not used
- Customer tier / segment -> not in your file; not used
- Line item ID -> not in your file; not used
- Product line name -> not in your file; not used
- Line category -> not in your file; not used
- List price of the line -> not in your file; not used
- Cost of the line -> not in your file; not used
- Discount on the line (%) -> not in your file; not used
- Was the line cancelled? (1/0) -> not in your file; not used
- Months until cancellation -> not in your file; not used

## Appendix: the last steps of the run

Every line was produced while the model ran. Re-running with the same seed replays it exactly.

| Time | Stage | What happened |
|---|---|---|
| 0.00s | prepare | Line category -> not in your file; not used |
| 0.00s | prepare | List price of the line -> not in your file; not used |
| 0.00s | prepare | Cost of the line -> not in your file; not used |
| 0.00s | prepare | Discount on the line (%) -> not in your file; not used |
| 0.00s | prepare | Was the line cancelled? (1/0) -> not in your file; not used |
| 0.00s | prepare | Months until cancellation -> not in your file; not used |
| 0.00s | prepare | Checking your assumptions… |
| 0.00s | simulate | Running the simulation… |
| 0.00s | read | Reading the quote you want us to stress-test... |
| 0.00s | read | The basket is worth $900k and carries $489k of profit (54.3% margin) if the buyer keeps every line. |
| 0.00s | read | The line being discounted hardest is Software at 35% off - that is 5 points past your 30% deep-discount line. |
| 0.00s | read | No past quote lines were supplied, so we will use the rates you typed in instead of learning from evidence. |
| 0.00s | prepare | Checking what happened in past deals that got a discount this deep... |
| 0.00s | prepare | Chance the offsetting lines disappear: 87.0% - that is what the discount is really buying. |
| 0.00s | simulate | Rebuilding 2,000 bundles that look like this one and re-pricing each of them... |
| 0.00s | simulate | Rebuilt 10% of the bundles... |
| 0.00s | simulate | Rebuilt 20% of the bundles... |
| 0.00s | simulate | Rebuilt 30% of the bundles... |
| 0.00s | simulate | Rebuilt 40% of the bundles... |
| 0.00s | simulate | Rebuilt 60% of the bundles... |
| 0.00s | simulate | Rebuilt 70% of the bundles... |
| 0.00s | simulate | Rebuilt 80% of the bundles... |
| 0.00s | simulate | Rebuilt 90% of the bundles... |
| 0.00s | summarize | All bundles priced. Working out what this deal really earns... |
| 0.00s | summarize | Done. Cherry-pick risk 71/100 (Moderate) - expect to lose $188k unless the discount is made conditional. |